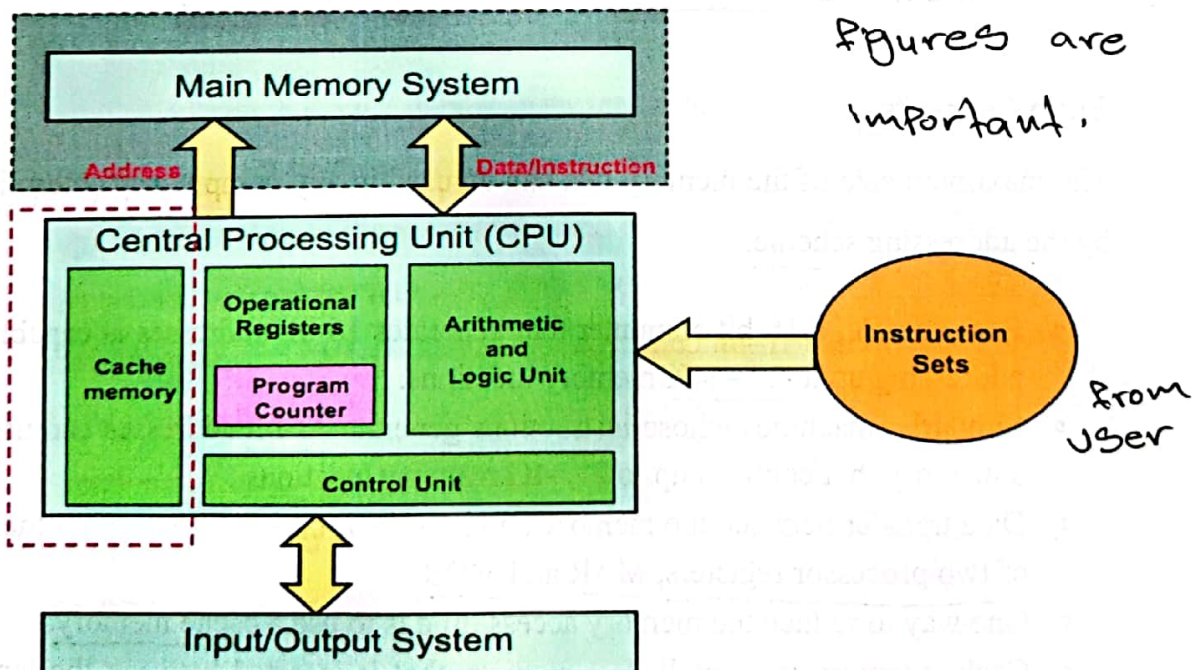


COMPUTER - MEMORY

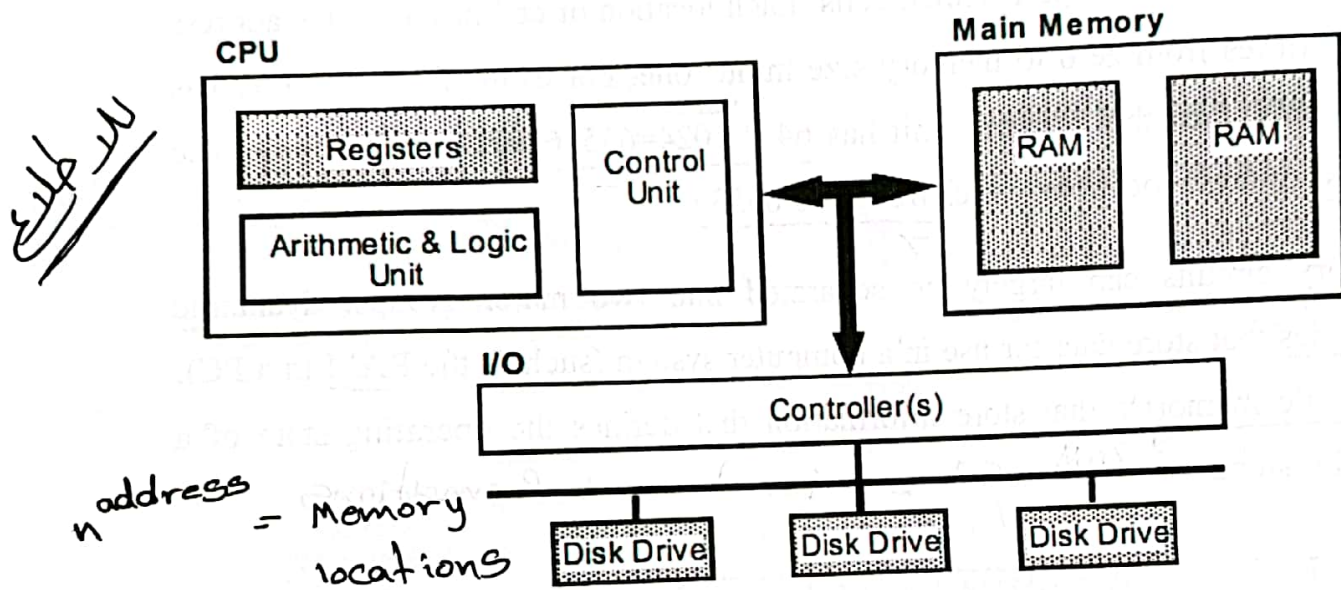
A memory is just like a human brain. It is used to store data and instructions. Computer memory is the storage space in computer where data is to be processed and instructions required for processing are stored. The memory is divided into large number of small parts called cells. Each location or cell has a unique address which varies from zero to memory size minus one. For example if computer has 64k words, then this memory unit has $64 * 1024 = 65536$ memory locations. The address of these locations varies from 0 to 65535.

Memory circuits can largely be separated into two major groups: dynamic memories that store data for use in a computer system (such as the RAM in a PC); and static memories that store information that defines the operating state of a digital system. $\rightarrow$ ROM Store operation informations



Main Memory (RAM) Organisation

Computers employ many different types of memory (semi-conductor, magnetic disks, USB sticks, DVDs etc.) to hold data and programs. Each type has its own characteristics and uses. We will look at the way that Main Memory (RAM) is organised and briefly at the characteristics of Register Memory and Disk Memory. Let us locate these 3 types of memory in a simplified model of a computer:



Basic Concepts

The maximum size of the memory that can be used in any computer is determined by the addressing scheme.

- For example, a 16-bit computer that generates 16-bit addresses is capable of addressing up to $2^{16} = 64K$ memory locations.
- Similarly, machines whose instructions generate 32-bit addresses can utilize a memory that contains up to $2^{32} = 4G$ memory locations.
- Data transfer between the memory and processor takes place through the use of two processor registers, MAR and MDR.
- One way to reduce the memory access time is to use a cache memory.
- Cache memory is a small, fast memory that is inserted between the larger, smaller main memory and the processor.

ROM → Read only memory

Read Right memory

~~Read Access memory~~

Main Memory (RAM) Random Access memory

If we were to sum all the bits of all registers within CPU, the total amount of memory probably would not exceed 5,000 bits. Most computational tasks undertaken by a computer require a lot more memory. Main memory is the next fastest memory within a computer and is much larger in size.

Typical main memory capacities for different kinds of computers are: PC 512MB, fileserver 4GB, database server 8GB.

Computer architectures also impose an architectural constraint on the maximum allowable RAM. This constraint is normally equal to 2^{WordSize} memory locations.

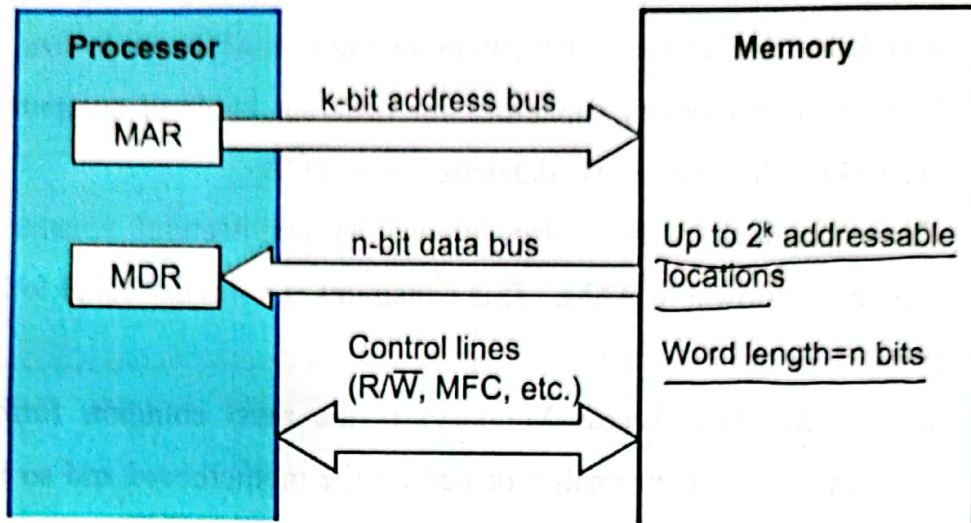
RAM (Random Access Memory) is the most common form of Main Memory. RAM is normally located on the motherboard and so is typically less than 12 inches from the CPU.

ROM (Read Only Memory) is like RAM except that its contents cannot be overwritten and its contents are not lost if power is turned off (ROM is non-volatile). Although slower than register memory, the contents of any location in RAM can still be "read" or "written" very quickly. The time to read or write is referred to as the access time and is constant for all RAM locations.

In contrast to register memory, RAM is used to hold both program code (instructions) and data (numbers, strings etc). Programs are "loaded" into RAM from a disk prior to execution by the CPU.

Locations in RAM are identified by an addressing scheme e.g. numbering the bytes in RAM from 0 onwards¹⁰. Like registers, the contents of RAM are lost if the power is turned off.

Connection of the Memory to the Processor



word = n bits

8 bits $\Rightarrow$ word length = 8

Memory is primarily of three types

- 1) Primary Memory/Main Memory.
- 2) Cache Memory .
- 3) Secondary Memory.

الفرق بين k -bit و n -bit

تنقل العناوين من دون تحديد طولها : k -bit
تنقل كميات فقط .

نقل البيانات بحجم أو طول البيانات : n -bit
المنقولة الموجودة والطول الخاص بال Word .

Primary Memory (Main Memory)

الذاكرة الرئيسية

Primary memory ^{تحتفظ} holds only those data and instructions on which computer is currently working. It has limited capacity and data is lost when power is switched off. It is generally made up of semiconductor device. These memories are not as fast as registers. The data and instruction required to be processed reside in main memory.

The processor will spend more of its time waiting to access the disk drive than carrying out program instructions. For this reason, the main memory where application and other support programs are loaded must have a speed comparable with that of the CPU itself. This unit is used for storing information (either data or programs), while being used in the computer system. Internal memories are designed for short-term, high-speed access of information. Internal memory also known as primary memory, main memory, or

simply memory.

نفس الأسماء لنفس الجزء

Internal memory (Primary memory) is divided into two subcategories RAM and ROM : **RAM** (Random Accesses Memory) and **ROM** (Read Only Memory).

2/2

1) Read-Write or Random-Access-Memory(RAM)

It has a variable content, also it is generally used to store the variable data-RAM can also be used to store frequently changed programs and other information. RAM allows the computer to store information quickly for later reference, so that (in most personal computer). RAM holds:-

RAM Plus

- The active part of the operating system, the fundamental program that control the operation of the computer.
- The application programs being executed (for example a word processing program).
- Data used by the application program (for example a letter being written with the word processing program).
- A representation of the data being presented on the video display.

A computer's memory is in constant communication with the CPU, as the program being executed. This type of memory contains a large number of semiconductor storage cells, each capable of storing a one-bit of instruction. A bit which is short of binary digit which is either 1 or 0 (either full or empty).

→ 8 bits
A group of eight bits is called a byte. Since a bite represents only a very small amount of information. The usual approach is to deal them in a groups of fixed size. For this purpose, the memory is organized so that a group of n-bits referred to as a word of information and n is called the word size (word length).

مسألة

Ex - Smart Home.

Memory chips:

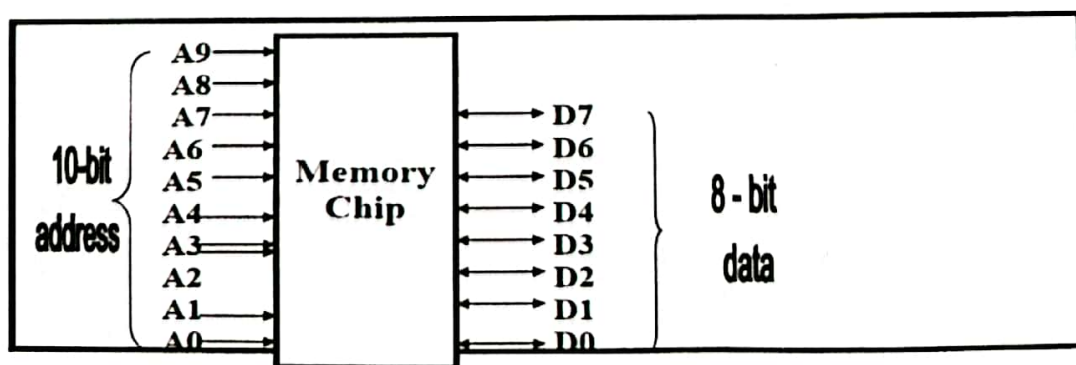
Memory chips have two main properties that determine their application, storage capacity or size and access time or speed.

A memory chip contains a number of locations, each of which stores one or more bits of data known as its **bit width**. The storage capacity of a memory chip is the product of the number of locations and the bit width. For example, a chip with 512 locations and a 2-bit data width, has a memory size of $512 \times 2 = 1024$ bits.

Since the standard unit of data is a byte(8 bits), the above storage capacity is normally given as $1024/8 = 128$ bytes.

Since the standard unit of data is a byte(8 bits), the above storage capacity is normally given as $1024/8 = 128$ bytes.

The number of locations may be obtained from the address width of the chip. For example, a chip with 10 address lines has $2^{10} = 1024$ or 1 k locations. Given an 8-bit data width, a 10-bit address chip has a memory size of $2^{10} \times 8 = 1024 \times 8 = 1k \times 8$ byte = 1 k byte or 1KB.



The computer's word size can be expressed in bytes as well as in bits.

For example, a word size of 8-bit is also a word size of one byte. a word size of 16-bit is a word size of two byte. Computers are often described in terms of their word size, such as an 8-bit computer, a 16-bit computer and so on.

For example, a 16-bit computer is one in which the instruction data are stored in memory as 16-bit units, and processed by the CPU in 16-bit units. The word size also indicates the size of the data. Bus which carries data between the CPU and memory and between the CPU and I/O devices.

To access the memory, to store or retrieve a single word of information, it is necessary to have a unique address. The word address is the number that identifies the location of a word in a memory.

Each word stored in a memory device has a unique address. Address are always expressed as binary number, although hexadecimal and decimal numbers are often used for convenience.

NOTS:

- ❖ The large computer(mainframes) have word-sizes that are usually in the 32-to-64 – bits range.
- ❖ Mini computers have a word sizes from 8-to-32-bits range.
- ❖ Microcomputers have a word sizes from 4-to-32-bits range.

In general a computer with a larger word size, can execute programs of instruction at a fast rate because more data and more instruction are stuffed into one word. The larger word sizes, however, mean more lines making up the data bus, and therefore more interconnections between the CPU and memory and I/O devices.

A more important than a computer's word size is the amount of memory the computer has, (i.e.) the memory capacity.

A memory capacity is a way of specifying how many bits can be stored in a particular memory device or complete memory system. The capacity of memory depends on two parameters, **the number of words (m)** and **the number of bits per word (n)**.

Memory capacity = (number of word) $\times$ (number of bits per word)

$$= m (\text{word}) * n (\text{bits})$$

$$= m * n \text{ bits}$$

EX:-

A certain memory chip is specified as $2K \times 8$:

1. How many words can be stored on this chip?
2. What is the words size?
3. How many total bits can this chip store?

SOL:-

1. $2K = 2 \times 1024 = 2048$ words
2. The word size is 8-bits (1 byte).
3. Capacity = $2048 \times 8 = 16,383$ bits = 16 KB.

EX:- A certain memory chip is specified as $2K \times 16$

1. How many words can be stored on this chip?
2. What is the words size?
3. How many total bits can this chip store?

Solution:-

1. $2K = 2 \times 1024 = 2048$ words
2. The word size is 16-bits(2 byte).
3. Capacity = $2048 * 16 = 32,768 = 23$ KB.